

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 8: JOURNEY INSIDE THE ATOM [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH08/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 In the Geiger-Marsden α -particle scattering experiment, what happens to the number of scattered α -particles detected as the scattering angle θ increases from 0° to 180° ?

- (A) It increases linearly (B) It drops dramatically according to $N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$
- (C) It remains constant at all angles (D) It drops to zero beyond 10°

Q.2 When an electron in a hydrogen atom transitions from a higher discrete energy orbit ($n = 3$) to a lower orbit ($n = 2$), what physical event occurs?

- (A) A proton is emitted from the nucleus (B) A photon of electromagnetic radiation with energy $\Delta E = h\nu$ is emitted
- (C) The atom collapses into a neutron (D) The electron is absorbed into the nucleus

Q.3 Which of the following isoelectronic ions has the smallest ionic radius: N^{3-} , O^{2-} , F^- , Na^+ , Mg^{2+} , Al^{3+} (all having 10 electrons)?

- (A) N^{3-} (B) O^{2-}
- (C) Na^+ (D) Al^{3+}

Q.4 Why does a neutron make an exceptionally effective projectile for inducing nuclear fission in heavy uranium-235 atoms?

- (A) Because neutrons carry positive charge (B) Because neutrons carry zero electrical charge and can approach the positively charged nucleus without experiencing electrostatic Coulomb repulsion
- (C) Because neutrons travel faster than light (D) Because neutrons have zero mass

Q.5 The electronic configuration of a divalent cation M^{2+} is 2, 8, 14. What is the atomic number of the neutral element M?

- (A) 24 (B) 26
- (C) 28 (D) 22

Q.6 What is the maximum number of electrons that can be accommodated in the M -shell ($n = 3$) of an atom?

- (A) 8 (B) 18
(C) 32 (D) 2

Q.7 Which of the following isotopes is utilized in industrial thickness gauges to monitor the thickness of manufactured plastic sheets and metal foils?

- (A) Cobalt-60 (B) Strontium-90 (beta emitter)
(C) Iodine-131 (D) Uranium-238

Q.8 An atom has 20 neutrons and mass number 39. What is its chemical valency?

- (A) 1 (B) 2
(C) 3 (D) 0

Q.9 Why did Dalton's original postulate that 'atoms are indivisible and indestructible' require scientific revision?

- (A) Because atoms turn into water (B) Because the discoveries of electrons, protons, and neutrons proved that atoms are composed of smaller subatomic particles
(C) Because atoms have no mass (D) Because all atoms are identical

Q.10 What is the mass ratio of a proton to an electron approximately?

- (A) 1 : 1 (B) 1840 : 1
(C) 1 : 1840 (D) 100 : 1

Q.11 An element X has an atomic number 15. What is the valency exhibited by X in the hydride XH_3 and chloride XCl_5 ?

- (A) 3 and 5 (B) 2 and 4
(C) 1 and 3 (D) Only 3

Q.12 Heavy water (D_2O) contains which isotope of hydrogen?

- (A) Protium (1_1H) (B) Deuterium (2_1H)
(C) Tritium (3_1H) (D) Helium

Q.13 Which of the following pairs represents isotones (species having identical number of neutrons)?

- (A) $^{14}_6C$ and $^{16}_8O$ (B) $^{12}_6C$ and $^{14}_6C$
(C) $^{40}_{18}Ar$ and $^{40}_{20}Ca$ (D) 1_1H and 2_1H

Q.14 Which subatomic particle was absent in Thomson's and Rutherford's early atomic models?

- (A) Electron (B) Proton
(C) Neutron (D) Positron

Q.15 What is the charge on an alpha (α) particle?

- (A) $+1$ unit (B) $+2$ units (equal to a helium nucleus He^{2+})
(C) -1 unit (D) Neutral

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): The chemical reactivity of an element is determined by its valence electrons.

Reason (R): In chemical reactions, only the outermost valence shell electrons are lost, gained, or shared to achieve stable octet configurations.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.17 Assertion (A): Rutherford's nuclear model could not explain the atomic emission spectra of hydrogen.

Reason (R): Classical electromagnetic theory predicted accelerating electrons would continuously radiate energy and collapse into the nucleus, leaving no explanation for discrete spectral lines.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.18 Assertion (A): The nucleus of an atom carries a positive electric charge.

Reason (R): The nucleus contains positively charged protons and electrically neutral neutrons.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.19 Assertion (A): Isotopes have identical physical properties such as boiling point and density.

Reason (R): Physical properties depend on mass, and isotopes of an element have different mass numbers.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.20 Assertion (A): Canal rays travel in straight lines towards the cathode.

Reason (R): Canal rays are produced by ionization of residual gas molecules in a discharge tube under high voltage.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Radioactive Isotope Tracers in Medicine & Industry

Radioactive isotopes (radioisotopes) emit ionizing radiation (alpha, beta, or gamma) as their unstable nuclei decay. In nuclear medicine and industry:

- **Technetium-99m:** Emits gamma rays with a 6-hour half-life used for imaging heart and bone perfusion.
- **Iodine-131:** Beta and gamma emitter with an 8-day half-life targeting the thyroid gland.
- **Cobalt-60:** Emits high-energy gamma rays (1.17 and 1.33 MeV) used in teletherapy cancer radiotherapy.
- **Americium-241:** Alpha emitter used in household smoke ionization detectors.

Q.21 Why is Technetium-99m ideal for diagnostic medical scans inside human patients?

- (A) It remains radioactive in the body for 50 years
 (B) Its short half-life of 6 hours allows clear gamma camera imaging while rapidly decaying to safe background levels, minimizing patient radiation dose
 (C) It is an alpha emitter that destroys all tissues
 (D) It is completely non-radioactive

Q.22 Why does radioactive Iodine-131 selectively accumulate in the human thyroid gland?

- (A) The thyroid gland naturally captures and concentrates iodine from the bloodstream to synthesize thyroid hormones (thyroxine)
 (B) Thyroid cells repel iodine
 (C) Iodine only binds to bone
 (D) Because of gastric acid

Q.23 In cancer radiotherapy, how do Cobalt-60 gamma rays destroy malignant cancer cells?

- (A) By freezing the tumor
 (B) High-energy gamma photons cause severe double-stranded DNA breaks in rapidly dividing cancer cells, triggering apoptosis
 (C) By starving cancer cells of oxygen
 (D) By turning cancer cells into healthy skin

Q.24 In a household smoke detector, why are alpha particles from Americium-241 utilized instead of gamma rays?

- (A) Alpha particles have high ionizing power to ionize air between detector plates, and smoke particles disrupt this current; their short range cannot penetrate the detector casing
 (B) Gamma rays are too heavy
 (C) Alpha particles are invisible
 (D) Smoke detectors require boiling water

Q.25 Which subatomic particle is identical to the alpha particle emitted by Americium-241?

- (A) A high-speed electron
 (B) A doubly ionized Helium-4 nucleus (${}^4\text{He}^{2+}$)
 (C) A neutral neutron
 (D) A positron

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(B)	Rutherford's scattering formula shows that the number of deflected particles drops inversely with $\sin^4(\theta/2)$; only a tiny fraction (1 in 12,000) experiences wide-angle backscattering near 180° .
Q.2	(B)	According to Bohr's model, when an electron jumps from a higher to lower stationary state, it emits a quantum photon of light whose energy equals the difference between the two orbital energy levels.
Q.3	(D)	All six ions have identical electron configurations (2, 8). Al^{3+} has the highest nuclear charge ($Z = 13$ protons), which exerts the strongest electrostatic pull on the 10 electrons, contracting the electron cloud to the smallest radius.
Q.4	(B)	Uncharged neutrons do not experience electrostatic repulsion from positive nuclear protons, allowing them to penetrate deep into heavy nuclei at low thermal speeds.
Q.5	(B)	Electrons in $\text{M}^{2+} = 2 + 8 + 14 = 24$. Since it lost 2 electrons, the neutral atom had $24 + 2 = 26$ electrons ($Z = 26$, Iron).
Q.6	(B)	$2n^2 = 2(3^2) = 2(9) = 18$ electrons.
Q.7	(B)	Beta emitters like Strontium-90 are absorbed proportional to material thickness, allowing automated radiation sensors to control roller pressure.
Q.8	(A)	$Z = A - N = 39 - 20 = 19$ (Potassium). Electronic configuration $= 2, 8, 8, 1$. Valency $= 1$.
Q.9	(B)	Subatomic particle discoveries (J.J. Thomson, Goldstein, Chadwick) and nuclear transmutation proved that atoms are divisible complex structures.
Q.10	(B)	Mass of proton $\approx 1.673 \times 10^{-27}$ kg; mass of electron $\approx 9.109 \times 10^{-31}$ kg. Ratio $\approx 1836 \approx 1840 : 1$.
Q.11	(A)	$Z = 15$ is Phosphorus (2, 8, 5). It exhibits variable valency 3 (gaining/sharing 3 electrons) and 5 (utilizing all 5 valence electrons with vacant d-orbitals).
Q.12	(B)	Heavy water is composed of deuterium (^2_1H), having one proton and one neutron, utilized as a moderator in nuclear reactors.
Q.13	(A)	In $^{14}_6\text{C}$, neutrons $= 14 - 6 = 8$. In $^{16}_8\text{O}$, neutrons $= 16 - 8 = 8$. Both have 8 neutrons (isotones).
Q.14	(C)	The neutron was not discovered until 1932 by James Chadwick, decades after Thomson (1897) and Rutherford (1911).
Q.15	(B)	An alpha particle consists of two protons and two neutrons, with a net charge of $+2e$ ($+3.2 \times 10^{-19}$ C) and mass ≈ 4 u.
Q.16	(A)	Core inner electrons are strongly shielded and tightly bound; chemical bonding and valency involve exclusively outer valence electrons.
Q.17	(A)	Continuous radiation would collapse the atom in 10^{-8} s and yield a continuous spectrum. Neils Bohr introduced discrete quantum orbits to solve this.
Q.18	(A)	Protons have charge $+e$ and neutrons have charge 0 , so the net nuclear charge equals $+Ze$.
Q.19	(D)	Assertion is false: Isotopes have <i>different</i> physical properties (density, boiling point, mass). Reason is true.
Q.20	(A)	Positively charged residual gas ions are accelerated towards the perforated negative cathode, passing through the canals as luminous streams.
Q.21	(B)	A 6-hour half-life gives sufficient time to complete diagnostic gamma scans while clearing quickly so the patient does not retain long-term radiation.
Q.22	(A)	The thyroid gland has active sodium-iodide symporters that concentrate iodine; localized beta radiation from ^{131}I destroys overactive thyroid or carcinoma cells.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.23	(B)	Ionizing radiation creates free radicals and direct DNA breaks; rapidly dividing cancer cells lack repair capacity and undergo mitotic death.
Q.24	(A)	Alpha particles strongly ionize air molecules between electrodes. Smoke absorbs alpha rays, dropping current and triggering the alarm. Alphas cannot penetrate the outer casing, posing zero hazard.
Q.25	(B)	An alpha particle consists of two protons and two neutrons tightly bound, identical to a helium-4 nucleus.